

The mapping class group action on the homology of the configuration spaces of a surface

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joint with Andrea Bianchi & Jeremy Miller

Algebraic structures in topology
16 July 2026

Today's goal:

Main Theorem (Miller–Bianchi–W)

$\Sigma_{g,1}$ = surface of genus g and 1 boundary component
 $F_n(\Sigma_{g,1})$ = its ordered configuration space on n points

Consider the action of the mapping class group on $H_i(F_n(\Sigma_{g,1}))$.

Let $g, n, i \geq 1$.

The i th term in the Johnson filtration $\mathcal{J}_{g,1}^i$ acts trivially on $H_i(F_n(\Sigma_{g,1}))$.

The topology of configuration spaces

Definition (configuration space)

M – topological space

$F_n(M)$ – (ordered) configuration space of M on n points

$$F_n(M) := \{(m_1, m_2, \dots, m_n) \in M^n \mid m_i \neq m_j \text{ for all } i \neq j\} \subseteq M^n$$

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Eg,  $\in F_4(M)$

The topology of configuration spaces

Definition (configuration space)

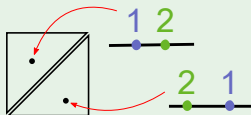
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$$\text{Eg, } \left(\begin{array}{c} \text{torus with 4 points} \\ \text{labeled } 1, 2, 3, 4 \end{array} \right) \in F_4(M)$$

Motivating example: configuration spaces

Definition (configuration space)

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$$\left\{ \begin{array}{c} \text{Connected components} \\ \text{of } F_n([0,1]) \end{array} \right\} \longleftrightarrow S_n$$



A horizontal line segment with four colored dots: green, blue, orange, and red. Above each dot is a number: 2 (green), 1 (blue), 4 (orange), and 3 (red).

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$F_n(D^2)$ is connected



An oval containing four colored dots: green, red, orange, and blue. Above each dot is a number: 2 (green), 3 (red), 4 (orange), and 1 (blue).

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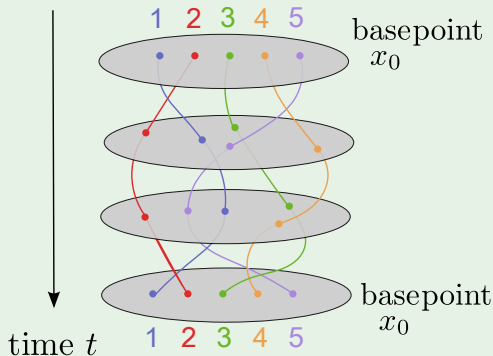
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$$\begin{array}{cccc} 2 & 1 & 4 & 3 \\ \bullet & \bullet & \bullet & \bullet \end{array} \in F_4([0,1])$$

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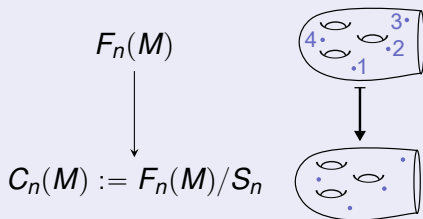
$$\pi_1(F_n(D^2)) = \text{Artin's pure braid group } \text{PB}_n$$



Unordered configuration spaces

$$S_n \curvearrowright F_n(M)$$

Definition (Unordered configuration space)



The *unordered configuration space* of M on n points is

$$C_n(M) = \left\{ \begin{array}{l} n\text{-element} \\ \text{subsets of } M \end{array} \right\}$$

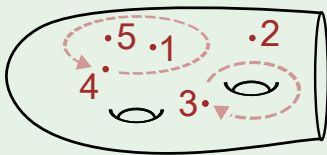
Homology of configuration spaces

Meta Goal: Understand $H_*(F_n(M))$ for M a d -manifold.

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$$S^1 \times S^1 \rightarrow F_5(M)$$

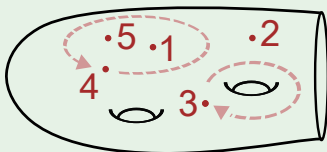


$\rightsquigarrow$ A class in
 $H_2(F_5(M))$
(up to sign)

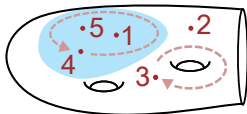
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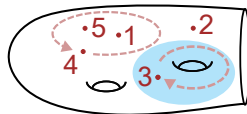
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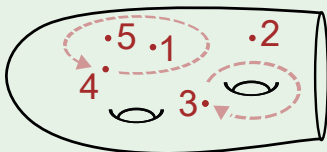


from $H_*(M)$

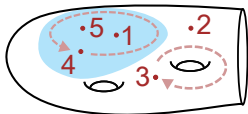
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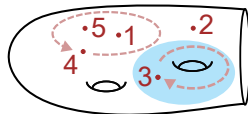
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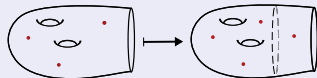
Hard problem: Understand additive relations between classes.

Classical Homological Stability for $C_n(M)$

M – connected, noncompact manifold of finite type, dimension $d \geq 2$

$\exists$ *stabilization map*

$$t : C_n(M) \rightarrow C_{n+1}(M)$$



Theorem (McDuff, Segal, 70s)

Fix M .

$\{C_n(M)\}_n$ is **homologically stable**.

For each i , the maps

$$t_* : H_i(C_n(M)) \rightarrow H_i(C_{n+1}(M))$$

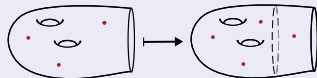
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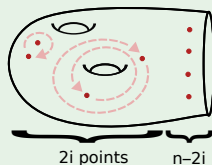
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$H_i(C_n(M))$ is spanned by:



Homological Stability for $F_n(M)$?

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Question: Is $\{F_n(M)\}_n$ homologically stable?

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$$H_1(F_n(D^2)) = \mathbb{Z}^{\binom{n}{2}},$$

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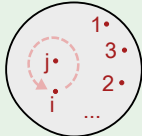

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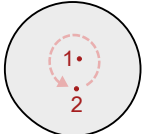
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Up to action of S_n
and stabilization map t ,
 $\{H_1(F_n(D^2))\}_n$ is generated by:

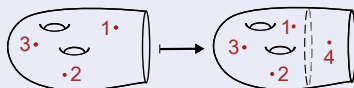
$$\alpha_{1,2} = \text{diagram} \in H_1(F_2(D^2))$$


Representation Stability for $F_n(M)$

M – connected, noncompact manifold of finite type,
dimension $d \geq 2$

$\exists$ stabilization map

$$t : F_n(M) \rightarrow F_{n+1}(M)$$

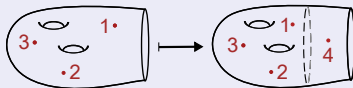


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Theorem (Church–Ellenberg–Farb, Miller–W (non-orientable M))

Fix M . For each fixed i , $\{H_i(F_n(M))\}_n$ is **representation stable**.

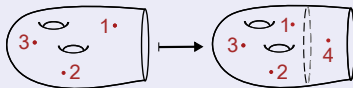
$$\mathbb{Z}[S_{n+1}] \cdot t_*(H_i(F_n(M); \mathbb{Z})) = H_i(F_{n+1}(M); \mathbb{Z}) \quad \text{for } n \geq 2i.$$

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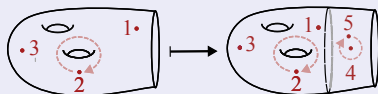


Secondary Stability for $F_n(M)$

M – connected, noncompact manifold of finite type,
dimension $d \geq 2$

$\exists$ stabilization map

$$t : H_i(F_n(M)) \rightarrow H_{i+1}(F_{n+2}(M))$$

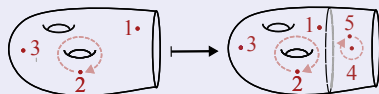


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Theorem (Miller-W)

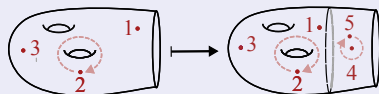
Fix M . $\{H_*(F_n(M))\}_n$ has **secondary representation stability**.

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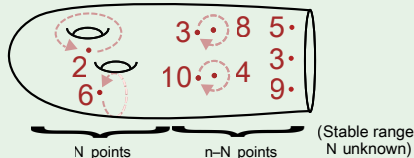
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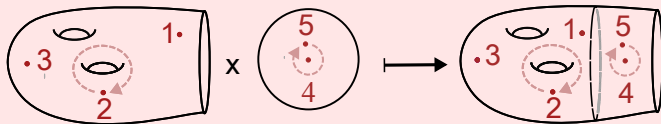
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Stability in the homology of configuration spaces

Philosophy:

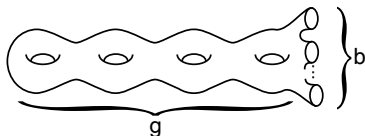
- View $H_*(F_\bullet(M))$ as a module over $H_*(F_\bullet(\mathbb{R}^d))$.



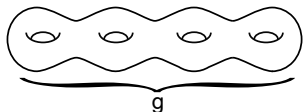
- The stability results are 'finite generation' results with respect to multiplication by fixed elements of $H_*(F_\bullet(\mathbb{R}^d))$.

Mapping class group actions

$\Sigma_{g,b}$ = compact orientable genus- g surface with b boundary components



$\Sigma_g = \Sigma_{g,0}$ = closed orientable genus- g surface

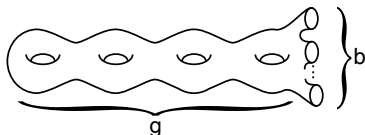


Definition (Mapping Class Group $\text{Mod}(\Sigma_{g,b})$)

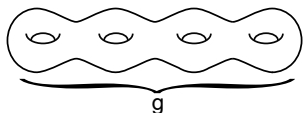
$\text{Mod}(\Sigma_{g,b}) := \text{Homeo}^+(\Sigma, \partial\Sigma_{g,b}) / (\text{isotopy stationary on } \partial\Sigma_{g,b})$

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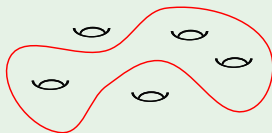
$$\text{Homeo}^+(\Sigma_{g,b}) \hookrightarrow F_n(\Sigma_{g,b}) \quad \rightsquigarrow \quad \text{Mod}(\Sigma_{g,b}) \hookrightarrow H_*(F_n(\Sigma_{g,b}))$$

Goal: What is the action of $\text{Mod}(\Sigma_{g,b})$ on $H_*(F_n(\Sigma_{g,b}))$?
What is its kernel?

The Mapping Class Group

Example (Dehn Twist about γ)

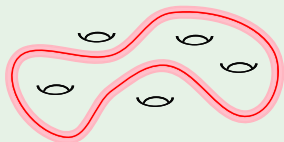
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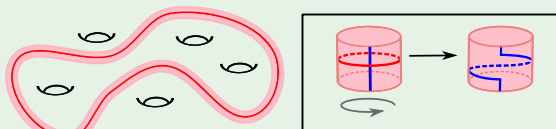


The Mapping Class Group

Example (Dehn Twist about γ)

γ – embedded loop in Σ

Dehn twist $T_\gamma : \Sigma \rightarrow \Sigma$

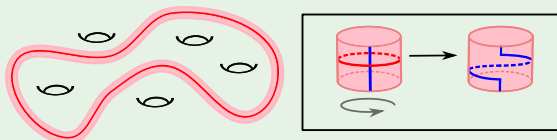


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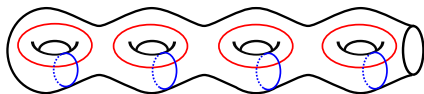
Theorem (Dehn, Mumford, Lickorish, Humphries)

$Mod(\Sigma_{g,b})$ is finitely generated by Dehn twists.

Action on Homology

Let $b \in \{0, 1\}$

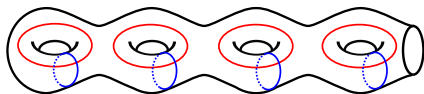
$$\text{Mod}(\Sigma_{g,b}) \curvearrowright H_1(\Sigma_{g,b}) \cong \mathbb{Z}^{2g}$$



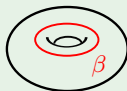
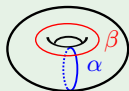
Action on Homology

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Example (Closed Torus)



$$\xrightarrow{T_\alpha}$$



$$T_\alpha(\beta) = \alpha + \beta$$

$$\text{Mod}(\Sigma_1) \xrightarrow{\cong} \text{Sp}_2(\mathbb{Z}) \cong \text{SL}_2(\mathbb{Z})$$

$$T_\alpha \mapsto \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$T_\beta \mapsto \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

The Torelli Group

Definition (Torelli group)

Let $b \in \{0, 1\}$.

$$\text{Torelli group } \mathcal{I}_{g,b} = \text{kernel} \left(\text{Mod}(\Sigma_{g,b}) \hookrightarrow H_1(\Sigma_{g,b}) \right)$$

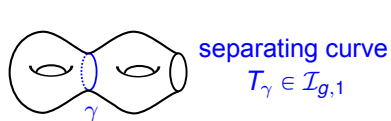
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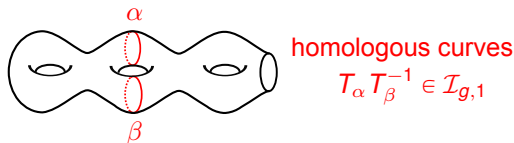
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Examples of mapping classes in Torelli:



“separating twist”

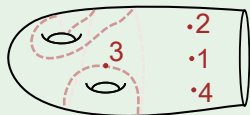


“bounding pair map”

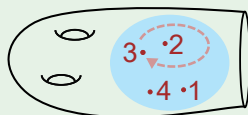
Action on degree-1 homology of configuration space

Example (Degree $i = 1$)

Let $n, g \geq 1$. $H_1(F_n(\Sigma_{g,b}))$ is generated by:



Loops arising from $H_1(\Sigma_{g,b})$

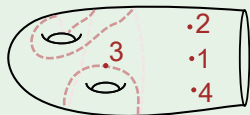


Loops arising from $H_1(F_n(\mathbb{R}^2))$

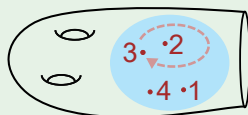
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Loops arising from $H_1(F_n(\mathbb{R}^2))$

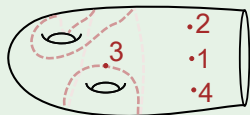
The Torelli group acts trivially on both types of classes.

Conclusion: $\mathcal{I}_{g,b} \subseteq \text{kernel} \left(\text{Mod}(\Sigma_{g,b}) \hookrightarrow H_1(F_n(\Sigma_{g,b})) \right)$

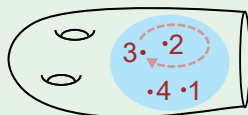
Action on degree-1 homology of configuration space

Example (Degree $i = 1$)

Let $n, g \geq 1$. $H_1(F_n(\Sigma_{g,b}))$ is generated by:



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Theorem (Bianchi)

Let $n, g \geq 2$. Then $\mathcal{I}_{g,1}$ acts **nontrivially** on $H_2(F_n(\Sigma_{g,1}))$.

The Johnson filtration

Choose basepoint $*$ $\in \partial \Sigma_{g,1}$. $\text{Mod}(\Sigma_{g,1})$ acts on $\pi := \pi_1(\Sigma_{g,1}, *)$.

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Definition (Johnson filtration)

Lower central series:

$$\gamma_0(\pi) = \pi$$

$\cup$

$$\gamma_1(\pi) = [\pi, \pi] \quad \begin{array}{l} \text{commutator} \\ \text{subgroup} \end{array}$$

$\cup$

$$\gamma_2(\pi) = [\pi, \gamma_1(\pi)]$$

$\cup$

$\vdots$

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Johnson filtration $\mathcal{J}_{g,1}^m =$

$$\text{kernel} \left(\text{Mod}(\Sigma_{g,1}) \hookrightarrow \pi / \gamma_m(\pi) \right)$$

$$\mathcal{J}_{g,1}^0 = \text{Mod}(\Sigma_{g,1})$$

$\cup$

$$\mathcal{J}_{g,1}^1 = \mathcal{I}_{g,1} \quad \text{Torelli subgroup}$$

$\cup$

$$\mathcal{J}_{g,1}^2 = \langle \text{separating twists} \rangle \quad \text{"Johnson kernel"}$$

$\cup$

$\vdots$

Main Theorems

Theorem (Moriyama)

Let $g, n, \geq 1$. Then $\mathcal{J}_{g,1}^n$ is the kernel of the action of $\text{Mod}(\Sigma_{g,1})$ on the compactly-supported cohomology group $H_c^(F_n(\Sigma_{g,1} \setminus *))$.*

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Let $g, n, i \geq 1$. Then $\mathcal{J}_{g,1}^i$ acts trivially on $H_i(F_n(\Sigma_{g,1}))$.

Proof: “Build” $H_*(F_\bullet(\Sigma_{g,1}))$ from $H_*(F_\bullet(\mathbb{R}^2))$ and Moriyama’s groups.

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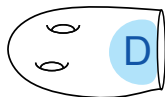
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$$\text{Moriyama's group} \cong H_n \left(\frac{F_n(\Sigma_{g,1})}{\text{configurations with at least 1 particle in } D} \right)$$



Encodes homology not arising from $F_n(\mathbb{R}^2)$

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Theorem (Bianchi–Stavrou): Main theorem is sharp

$\mathcal{J}_{g,1}^{n-1}$ acts **nontrivially** on $H_n(F_n(\Sigma_{g,1}))$ for $g \geq 2$ and $n \geq 1$.

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Stavrou: description of action for **unordered** configurations

Looijenga–Stavrou: generalizations to CW cplx including Σ_g and $\Sigma_{g,b}$

Thank you!

